

Counting

CSE106
East West University

Addition Rule

For two separate tasks A and B

If A can be done in m ways, and
B can be done in n ways

Then A and B can be done in $m+n$ ways

How many ways can you roll an even number on a die or draw a face card ?

A

Die

2

4

6

3

+

12

Total 15

B

Card

♥J ♥Q ♥K

♦J ♦Q ♦K

♠J ♠Q ♠K

♣J ♣Q ♣K

Basic Counting Principles

The sum rule:

• If a task can be done in n_1 ways and a second task in n_2 ways, and if these two tasks cannot be done at the same time, then there $n_1 + n_2$ ways to do either task.

Example:

• The department will award a free computer to either a CS student or a CS Professor. How many different choices are there, if there are 530 students and 15 professors?

Ans: $530 + 15 \rightarrow 545$

Example

- A student can choose a computer project from one of three lists. The three lists contain 23, 15, and 19 possible projects, respectively. No project is on more than one list. How many possible projects are there to choose from ?

$$\begin{array}{ccc} \underline{23} & + & \underline{15} & + & \underline{19} \\ A & & B & & C \end{array}$$

• **Sol:**

The student can choose a project by selecting a project from the first list, the second list, or the third list. Because no project is on more than one list, by the sum rule there are

$23 + 15 + 19 = 57$ ways to choose a project.

Arriving at the rule of Sum

If we roll two dice, one green and one purple, how many ways are there to get a sum of 7 or 11?

(1,6)

(2,5)

First let's look at a sum of 7: (3,4)

(1,6) (6,1) (2,5) (5,2) (3,4) (4,3)

And now a sum of 11:

(5,6) (6,5)

Since there are 6 ways to get a sum of 7, and 2 ways to get a sum of 11, we have $6 + 2 = 8$ ways to get a sum of 7 or 11.

The Rule of Sum One event , multiple ways

If a task can be done in either one of n_1 ways **or** n_2 ways, where two sets of ways are **disjoint**, then there are $n_1 + n_2$ ways to complete the task.

During a local campaign, three Republican and two Democratic candidates are nominated for president of the school board. If the president is to be one of these candidates, how many possibilities are there for an eventual winner? $3+2 = 5$

Rule of Product

In subsequent tasks, A and B,

If there are m ways to do task A, and

n ways to do task B,

Then there are mn ways to do task A then B.

Basic Counting Principles

The product rule:

- Suppose that a procedure can be broken down into a sequence of two tasks.
- If there are n_1 ways to do the first task and for each of these ways doing the first task, there are n_2 ways to do the second task,
- Then there are $n_1 n_2$ ways to do the procedure.

Example:

- How many different license plates are there that containing exactly three English letters ?

Solution:

There are 26 possibilities to pick the first letter, then 26 possibilities for the second one and 26 for the last one.

- So there are $26 \cdot 26 \cdot 26 = 17576$ different license plates.

Example

- A new company with just two employees, Ali and Ahmed, rents a floor of a building with 12 offices. How many ways are there to assign different offices to these two employees?

Solve:

- The procedure of assigning offices to these two employees consists of assigning an office to Ali, which can be done in 12 ways.
- then assigning an office to Ahmed different from the office assigned to Ali, which can be done in 11 ways.
- By the product rule, there are $12 * 11 = 132$ ways to assign offices to these two employees.

This above example is product because cannot assign them rooms separately. Have to assign a room to one employee, and then from the remaining rooms, assign one to the other worker.

Example

- The chairs of an auditorium are to be labeled with an uppercase English letter followed by a positive integer not exceeding 100. What is the largest number of chairs that can be labeled differently ?

Solve:

- The procedure of labeling a chair consists of two tasks, namely, assigning to the seat one of the 26 uppercase English letters,
- Then assigning to one of the 100 possible integers.
- The product rule shows that there are $26 * 100 = 2600$ different ways that a chair can be labeled.

Examples

1. There are **32** microcomputers in a computer center. Each microcomputer has **24** ports. How many different ports to a microcomputer in the center are there ?

$$32 * 24 = 748$$

2. How many different bit strings of length **10** seven are there ?

$$2 * 2 * 2 * 2 * 2 * 2 * 2 * 2 = 2^7 = 128$$

3. How many different license plates can be made if each plate contains a sequence of three uppercase English letters followed by three digits (and no sequences of letters are prohibited)?

$$\begin{array}{c} \underline{\text{A-Z}} \quad \underline{\text{A-Z}} \quad \underline{\text{A-Z}} \\ 26 * 26 * 26 * 10 * 10 * 10 = 17,576,000 \end{array}$$

4. How many functions are there from a set with m elements to a set of n elements? **n^m**

Examples solution

In 2), there are two bits 0 and 1. And this two combinations will be in 7 length. So, 2^7

In 4), m to n is the function, so m is the domain and n is the codomain. To be a function, each m has to have a value. Therefore, space will be of length m and values will be n . Answer: $n \times n \times n \dots$ this will be n^m . Like in 2)

Example

- Suppose that either a member of the mathematics faculty or a student who is a mathematics major is chosen as a representative to a university committee. How many different choices are there for this representative if there are 37 members of the mathematics faculty and 83 mathematics majors and no one is both a faculty member and a student ?

Solve:

- Which rule is applicable here i.e; sum or product ?
- There are 37 ways to choose a member of the mathematics faculty and there are 83 ways to choose a student who is a mathematics major.
- Choosing a member of the mathematics faculty is never the same as choosing a student who is a mathematics major because no one is both a faculty member and a student.

By the sum rule it follows that there are $37 + 83 = 120$ possible ways to pick this representative

Sandwich Example

I want to make a sandwich with one thing from each category.

- A. Top Bread:** White, Whole Wheat, Sour Dough $\rightarrow 3$
- B. Meat:** Deshi Chicken, Mutton, Wagyu, Beef Steak, Farm Chicken, Sheep, Duck $\rightarrow 7$
- C. Vegetable:** Lettuce, Tomato, Cucumber, Turnip $\rightarrow 4$
- D. Bottom Bread:** White, Whole Wheat, Sour Dough $\rightarrow 3$

How many ways can i do that?

$$3 \times 7 \times 4 \times 3 = 21 \times 12 = 252 \text{ ways}$$

License Plate Example

How many ways can we make a license plate with ...

(a) 3 even numbers and 3 letters

$$5 \times 5 \times 5 \times 26 \times 26 \times 26 = 2197000 \text{ ways}$$

(b) 2 numbers, 2 letters, 1 odd number, 1 even number, and 2 vowels?

$$10 \times 10 \times 26 \times 26 \times 5 \times 5 \times 5 \times 5 = 42250000 \text{ ways}$$

(c) 6 letters that are not the same?

$$26 \times 25 \times 24 \times 23 \times 22 \times 21 = 165765600 \text{ ways}$$

President Example

8 women and 5 men are elected for president.

I. How many ways I chose a winner?

$$8+5=13$$

II. How many man vs woman pair can be made?

$$5 \times 8 = 40$$

III. How many ways can we choose a president and a female vice president?

$$(5 \times 8) + (8 \times 7) = 96$$

Another Sandwich Example

Want to buy a sandwich.

- 3 bread
- 5 meat
- 6 cheese

(1) How many ways to make a 1 meat, 1 cheese sandwich?

$$3 \times 5 \times 6 = 90 \text{ ways}$$

(2) How many ways to make a 3 meat, 2 cheese sandwich? [No repeat]

$$3 \times (5 \times 4 \times 3) \times (6 \times 5) = 5400 \text{ ways}$$

We have 3 bread choices.

Then as there are no meat repeats = 5 meat $\times$ 4 meat $\times$ 3 meat.

Also, as no cheese repeat = 6 cheese $\times$ 5 cheese.

Practice

A student ID is made up of 3 uppercase letters followed by two digits.

1. How many possible IDs exist?

$$26 \times 26 \times 26 \times 10 \times 10 = 26^3 \times 10^2 = 1757600$$

2. How many IDs are possible if duplicate letters or numbers are not allowed?

$$26 \times 25 \times 24 \times 10 \times 9 = 1404000$$

3. How many student IDs are possible with an even number of “A”s?

$$25 \times 25 \times 25 \times 10 \times 10 = 1562500$$

$$1562500 + (3 \times 2500) = 1570000$$

Practice

Here, in part 3, Even number of A's in 3 Upper case letters.

So either 0 "A"s or 2 "A"s are the options for having even "A"s.

- For zero "A"s, we will have no A so 25 alphabets.

- For 2 "A"s, we will have 3 combinations of how the two As will be positioned in the 3 locations provided for upper case letters-
which results to $= (25 \times 100) \times 3 = 7500$

These two are the combinations so we summate.

Example 7

Counting One-to-One Functions How many one-to-one functions are there from a set with m elements to one with n elements?

Counting the number of onto functions is harder. We'll do this in Chapter 8.

Solution: First note that when $m > n$ there are no one-to-one functions from a set with m elements to a set with n elements.

Now let $m \leq n$. Suppose the elements in the domain are $a_1, a_2, \dots, a_m$. There are n ways to choose the value of the function at a_1 . Because the function is one-to-one, the value of the function at a_2 can be picked in $n - 1$ ways (because the value used for a_1 cannot be used again).

In general, the value of the function at a_k can be chosen in $(n - k + 1)$ ways. By the product rule, there are $n(n-1)(n-2)\dots(n-m+1)$ one-to-one functions from a set with m elements to one with n elements.

For example, there are $5 \times 4 \times 3 = 60$ one-to-one functions from a set with three elements to a set with five elements.

COMPLICATED EXAMPLES: Example 15

Many counting problems cannot be solved using just the sum rule or just the product rule. However, many complicated counting problems can be solved using both of these rules in combination. We begin by counting the number of variable names in the programming language BASIC. (In the exercises, we consider the number of variable names in JAVA.) Then we will count the number of valid passwords subject to a particular set of restrictions.

EXAMPLE 15 In a version of the computer language BASIC, the name of a variable is a string of one or two alphanumeric characters, where uppercase and lowercase letters are not distinguished. (An alphanumeric character is either one of the 26 English letters or one of the 10 digits.) Moreover, a variable name must begin with a letter and must be different from the five strings of two characters that are reserved for programming use. How many different variable names are there in this version of BASIC?

Solution of Example 15

Solution: Let V equal the number of different variable names in this version of BASIC. Let V_1 be the number of these that are one character long and V_2 be the number of these that are two characters long. Then by the sum rule, $V = V_1 + V_2$. Note that $V_1 = 26$, because a one-character variable name must be a letter. Furthermore, by the product rule there are $26 \cdot 36$ strings of length two that begin with a letter and end with an alphanumeric character. However, five of these are excluded, so $V_2 = 26 \cdot 36 - 5 = 931$. Hence, there are $V = V_1 + V_2 = 26 + 931 = 957$ different names for variables in this version of BASIC.

BASIC

BASIC is a programming language. Here, the variable naming template can be one or two digits consisting of alphanumeric characters. And cannot start with number, has to be alphabet. So there will be two combinations: one digit or two digit.

V1 is one digit and so as cannot start with number, will be alphabet.

V2 is two digit, so will start with alphabet - 26, and the 2nd digit will have alphanumeric- 26 alphabet and 10 digits (0-9) so 36.

Another clause is that the name CANNOT be the same as the five strings that are used for programming in BASIC.

So V2 is $(26 \times 36) - 5 = 931$

$V = V1 + V2$

Subtraction Rule

INCLUSION EXCLUSION:

When two activities are independent of each other, BUT have ways that are common between them, then we cannot use addition rule. The reason for this is that, if we add, then the common way between them will be added twice-overcount.

To correctly count the number of ways to do the two tasks, we must subtract the number of ways that are counted twice. This leads us to an important counting rule.

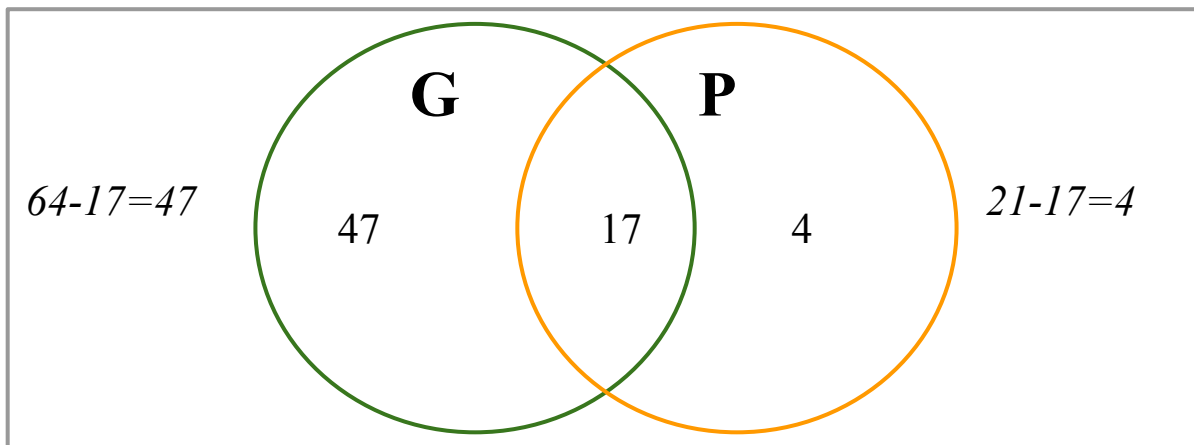
THE SUBTRACTION RULE :

If a task can be done in either n_1 ways or n_2 ways, then the number of ways to do the task is $n_1 + n_2$ minus the number of ways to do the task that are common to the two different ways.

Venn Diagram Example

In the real estate ads, 64 homes for sale have garages, 21 have swimming pools, and 17 have both a garage and a pool. How many total homes are for sale?

$$47+17+4=68$$



Venn Diagram to Formula

In the above example, if we draw a venn diagram, the situation and counting can be mapped properly. But it is tiresome to draw a venn diagram all the time for this.

So, the formula:

$$|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$$

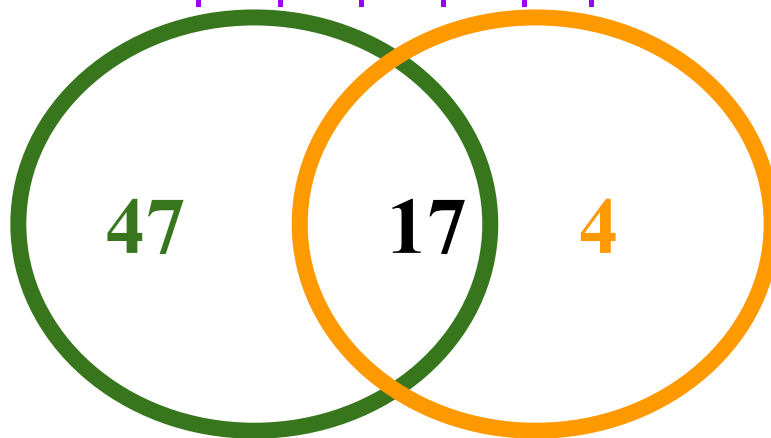
How the formula works:

Adding the ways of both A1 and A2 will give us the common way between them twice. So, by subtracting $A_1 \cap A_2$, we remove one of the common ways that were added twice, making the count of the common way one.

The Principle of Inclusion-Exclusion

The number of elements in the union of two finite sets, A and B, can be modeled by:

$$|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$$



Here, we count A once, B once, and in their intersection the common value is counted twice. To remove one, we subtract $A \cap B$ once.

Another Example

If there are only computer science and math majors in discrete math, and there are 42 computer science majors, 15 math majors, and 8 student who are double majors in math and computer science, how many people are enrolled in discrete math?

$$\begin{aligned} |\mathbf{M} \cup \mathbf{C}| &= |\mathbf{M}| + |\mathbf{C}| - |\mathbf{M} \cap \mathbf{C}| \\ &= 15 + 42 - 8 \\ &= 49 \end{aligned}$$

Example 18

How many bit strings of length eight either start with a 1 bit or end with the two bits 00?

Solution: Figure 2 illustrates the three counting problems we need to solve before we can apply the principle of inclusion-exclusion. We can construct a bit string of length eight that either starts with a 1 bit or ends with the two bits 00, by constructing a bit string of length eight beginning with a 1 bit or by constructing a bit string of length eight that ends with the two bits 00. We can construct a bit string of length eight that begins with a 1 in $2^7 = 128$ ways. This follows by the product rule, because the first bit can be chosen in only one way and each of the other seven bits can be chosen in two ways. Similarly, we can construct a bit string of length eight ending with the two bits 00, in $2^6 = 64$ ways. This follows by the product rule, because each of the first six bits can be chosen in two ways and the last two bits can be chosen in only one way.

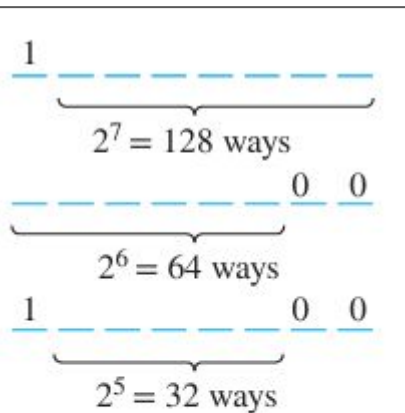


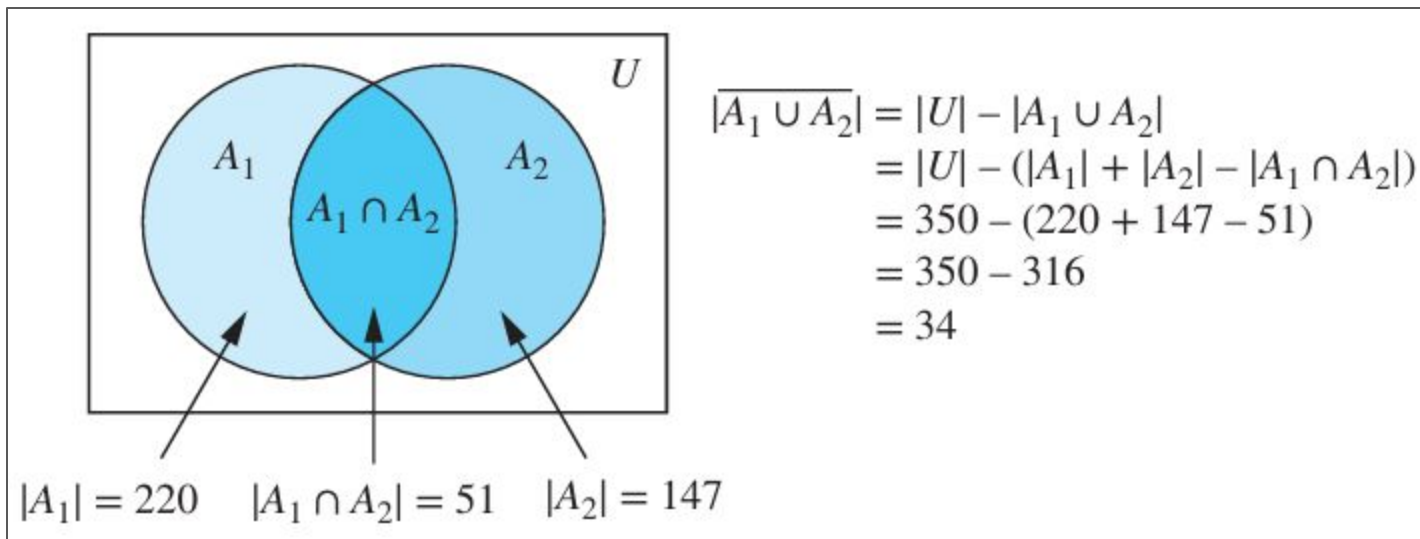
FIGURE 2 8-Bit strings starting with 1 or ending with 00.

Example 18 (continues)

Some of the ways to construct a bit string of length eight starting with a 1 are the same as the ways to construct a bit string of length eight that ends with the two bits 00. There are $2^5=32$ ways to construct such a string. This follows by the product rule, because the first bit can be chosen in only one way, each of the second through the sixth bits can be chosen in two ways, and the last two bits can be chosen in one way. Consequently, the number of bit strings of length eight that begin with a 1 or end with a 00, which equals the number of ways to construct a bit string of length eight that begins with a 1 or that ends with 00, equals $128+64-32 = 160$.

Example 19

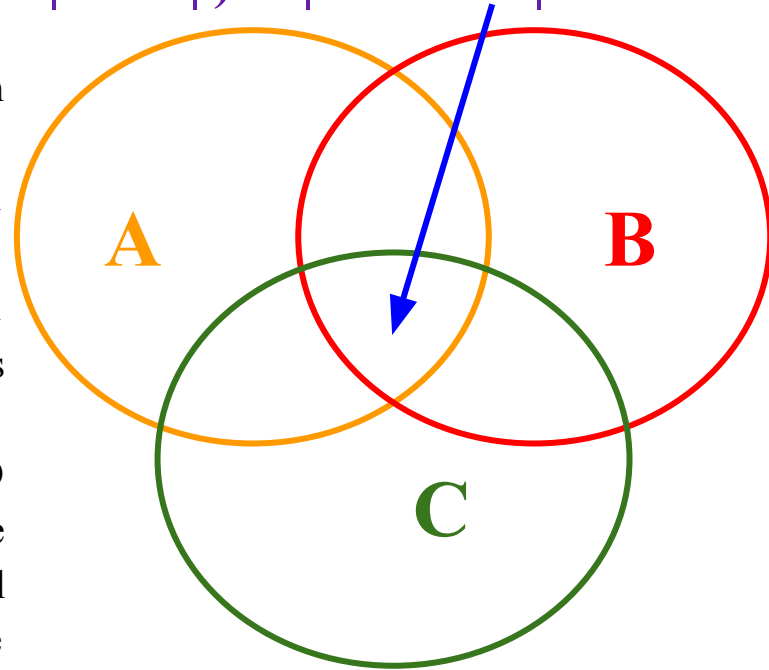
A computer company receives 350 applications from college graduates for a job planning a line of new web servers. Suppose that 220 of these applicants majored in computer science, 147 majored in business, and 51 majored both in computer science and in business. How many of these applicants majored neither in computer science nor in business?



Inclusion-Exclusion For 3 Sets

$$|A \cup B \cup C| = |A| + |B| + |C| - (|A \cap B| + |A \cap C| + |B \cap C|) + |A \cap B \cap C|$$

When there are THREE sets and they have commons between them, we add each set then subtract $A \cap B$, $B \cap C$, and $A \cap C$. When we subtract $A \cap B$ from A , one count is removed from these, and one which BELONGS TO B remains as this is $A \cap B$, not $B \cap A$. Another intersection between A and B is in the common area between all three sets. So we subtract one count from here as well, and two remains. Similarly, for $A \cap C$ and $B \cap C$ counts will be subtracted in such a way that the intersections between TWO set combination will have one value in it(which is fine), but the intersection between ALL THREE will be 0. All the 3 counts will be subtracted. Therefore, we add $A \cap B \cap C$ at the end of the equation as given below.



Question

A large software development company employs 100 computer programmers. Of them, 45 are proficient in Java, 30 in C#, 20 in Python, six in C# and Java, one in Java and Python, five in C# and Python, and just one programmer is proficient in all three languages above. Determine the number of computer programmers that are not proficient in any of these three languages.

$$|U| = 100$$

$$|J| = 45, \quad |C| = 30, \quad |P| = 20$$

$$|J \cap C| = 6, \quad |J \cap P| = 1, \quad |C \cap P| = 5$$

$$|J \cap C \cap P| = 1$$

$$\text{Answer} = |U| - |J \cup C \cup P|$$

$$= 100 - (45 + 30 + 20 - (6 + 1 + 5) + 1) = 16$$

16 programmers are not proficient in any of the three languages.

Tree Diagram will be studied in later chapters, so for this chapter we will skip it.

PIGEONHOLE THEOREM (For Later)

In mathematics, the pigeonhole principle states that if n items are put into m containers, with $n > m$, then at least one container must contain more than one item.

If k is a positive integer and $k + 1$ or more objects are placed into k boxes, then there is at least one box containing two or more (at least two) of the objects.

Example 1

Among any group of 367 people, there must be at least two with the same birthday, because there are only 366 possible birthdays.

Counting February 29, which comes around every four years on Leap Day, there are 366 possible days you could be born.

If N objects are placed into k boxes, then there is at least one box containing at least $r = \lceil N / k \rceil$ objects, where $\lceil \cdot \rceil$ is the ceiling value.

5 holes, 6 pigeons. So $\lceil 6/5 \rceil = 2$ as ceiling. So there will be at least one box which will have 2 pigeons in it.

Generalized Pigeonhole Principle

If N objects are placed into k boxes, then there is at least one box containing $\lceil N / k \rceil$ objects.

Among 50 people, what is the number of people that must be born on the same month?

$$\lceil 50 / 12 \rceil = \lceil 4.16 \rceil = 5$$